Inhibition of Return and Learned Value

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Abstract

Salient stimuli automatically attract attention, leading to facilitation effects, but later reduce the likelihood that attention will be recaptured at the same location or by the same object, a phenomenon known as inhibition of return (IOR). Interestingly, IOR is a modulable phenomenon in response to different manipulations of both cue and target. To date, few studies have investigated whether the arbitrary association of an irrelevant dimension (e.g., color) with different levels of reward can affect IOR. In two pre-registered studies, we manipulated the amount of reward associated with a target in a Posner-type task and examined whether such a manipulation affected the IOR effect in a phase in which the participant received feedback on his performance and in a phase in which feedback was omitted. Experiment 1 showed that learned value can modulate the IOR effect only when feedback is omitted. Experiment 2 nominally replicated the previous results, but not statistically. Suspecting a relatively small effect size, we analyzed both experiments together (total $N = 100$) observing a consistent reduction of the IOR effect for targets associated with high reward compared to those associated with low reward. A meta-analysis of the current and two other conceptually similar published studies further suggests that although value can modulate IOR, the effect size may be small, and a high sample size may be needed to research this topic. These results are discussed in light of recent theories of attentional orienting and return inhibition.

Keywords: Learned value, Selection history, Inhibition of Return, Spatial Cueing

Public Significance Statement

Our experience of the environment shapes our attention in many ways. Although it is known that objects that predict reward are more salient than others, the extent to which stimulus-reward associations can influence our ability to orient attention is unknown. This study shows that reward-predicting stimuli can modulate inhibition of return. This finding contradicts classical reorientation theories of inhibition of return and extends what we know about how learning certain reward-predicting properties can affect our ability to orient attention.

Introduction

Imagine you need to find something. If you are guided by a "cue" that tells you where to look, you will find it more quickly. In fact, in the field of attentional orienting, we can distinguish between two types of orienting: (1) top-down, "voluntary" or endogenous orienting (orienting attention according to our goals, intentions, or task demands); and (2) bottom-up, "involuntary" or exogenous orienting (orienting attention to potentially relevant stimuli) (Jonides & Irwin, 1981). A paradigm that allows the study of attentional orienting and the dissociation between these two types of orienting is the "costs and benefits" paradigm by Posner and Cohen (1984). In this paradigm, a target is presented to the left or right of fixation and participants are asked to respond as quickly and accurately as possible. Different spatial cues preceding the target and indicating its location can be used to investigate endogenous and exogenous attentional orienting. Central cues predicting the target location are used to investigate endogenous, rather voluntary attention. Compared to a neutral condition, responses are faster at the location predicted by the cue (valid trials), leading to the so-called attentional benefit effect, and slower at the opposite, invalid location, leading to the so-called attentional cost effect. To investigate exogenous, more automatic attention, peripheral cues are used to cue the appearance of the target, but critically, the cues are not predictive of the target location, so that in 50% of the trials the cues indicate where the target will appear (cued location trials) and in the other 50% the cue appears at the opposite location (uncued location trials). Although the observed effects depend on the task set and other factors (Chica et al, 2014), the typical pattern of results consists of faster RTs for cued than uncued location trials (facilitation effect) when stimulus onset asynchrony (SOA) is short (typically less than about 200 ms); conversely, for larger SOAs, RT for cued location trials is slower than that for uncued location trials, an effect Posner and colleagues (1985) called

inhibition of return (IOR) (although see Berlucchi, 2006, for a discussion of the inappropriateness of the name).

On the other hand, we will find what is important to us more quickly. Indeed, the interplay between attention and motivation is an important determinant of performance in many tasks (Esterman et al., 2016; Padmala & Pessoa, 2011). The effect of motivation on attention is often explained by an improvement in cognitive control mechanisms (Botvinick & Braver, 2015). Therefore, when trying to predict how motivation modulates attentional orienting, it is reasonable to expect that motivation will improve endogenous rather than exogenous orienting. To demonstrate the interaction between motivation and attentional orienting, Engelmann and Pessoa (2014) used a detection task from a spatial cueing paradigm with peripheral and predictive cues to investigate how economic gains and losses interact with spatial orienting. In their task, the background color signaled whether participants' performance would lead to the possibility of gaining money or avoiding losing money. In this setting, Engelmann and Pessoa (2014) found that either reward or punishment produced an increase in performance regardless of the validity of the cue. That is, reward-induced motivation produces an increase in performance in exogenous orienting and reorienting processes. Although the lack of interaction could be explained by the fact that motivation similarly affects orienting (on valid trials, thus increasing attentional benefits or the attentional facilitation effect) and reorienting processes (on invalid trials, thus reducing attentional costs), one difficulty in concluding that reward modulates exogenous orienting is that although the type of cue affects the type of orientation we measure, so does its predictability (Lupiáñez et al., 2004). Thus, the use of peripheral informative cues does not allow us to properly dissociate exogenous from endogenous orienting processes.

One of the first studies attempting to investigate the effect of reward on a completely exogenous orienting task was proposed by Bucker and Theeuwes (2014). In their work, Bucker and Theeuwes (2014) used a discrimination task based on the spatial cueing paradigm described above. They used two SOAs (170 ms and 970 ms) and alternating blocks in which participants had the opportunity to receive a high or low reward ($0.1 or $1). The results observed by Bucker and Theeuwes (2014) show that while the value of the reward does not affect RTs in the short SOA, it does in the long SOA, with the IOR effect disappearing in the low-reward blocks. Bucker and Theeuwes (2014) argue that this pattern is explained by the fact that the administration of monetary rewards does not affect the initial orientation of attention, which is purely exogenous, while it does affect IOR since the inhibition of the previously attended location depends on top-down processes. In light of these findings, they conclude that performance-based rewards influence endogenous but not exogenous orienting.

Thus, while rewarding performance generally influences the participant's motivational state, contingently associating a stimulus feature with reward may affect more specific processes. For example, this specific association appears to alter the ability of that stimulus feature to capture attention (Anderson et al., 2011; Della Libera & Chelazzi, 2009), even when the stimulus predicting reward acts as an irrelevant distractor so that attending to it is completely counterproductive (Le Pelley et al., 2015; Le Pelley et al., 2017), or when reward is no longer available to motivate performance (Anderson & Halpern, 2017; Anderson et al., 2011; Watson et al., 2019; Garre-Frutos et al., 2024). Thus, this bias does not consider endogenous factors or the intrinsic capacity of the stimulus to capture attention and has therefore been proposed as an effect belonging to the so-called selection history (Anderson et al., 2021) that involuntarily influences attentional orienting.

In the above study, Bucker and Theeuwes (2014) investigated how performance-based reward affects attentional orienting from a spatial cueing paradigm, but they did not show evidence of how stimulus-reward association could affect orienting. Some studies have tried to disentangle how color-reward association could interact with attentional orienting by using cueing paradigms. For example, Bourgeois et al. (2016) used a training phase in which they created color-reward associations (high and low-value stimulus) and then used the same stimulus in a testing phase as distractors in a visual search task, in which color was irrelevant and the target was defined by shape (a diamond among circles). Critically, the testing phase used an endogenous or an exogenous cue to signal the target position, finding that a color distractor previously associated with a high value impairs performance in all conditions, irrespective of cue validity and exogenous or endogenous attention. Thus, in their study, Bourgeois et al. (2016) interpreted that the previously rewarded color distractor captures attention affecting the reorienting of attention to the target in both exogenous and endogenous cueing paradigms.

Moreover, Failing and Theeuwes (2014) proved that a cue-signaling reward could capture attention independently of cue salience. In their study, Failing and Theeuwes (2014) used a training phase in which they present concurrently two circles out of four possible. The participant had to discriminate between two target letters that could only appear inside two possible circle colors: one rewarded and the other non-rewarded (but “trained”). Then in a test phase with no reward, Failing and Theeuwes (2014) found that the rewarded stimulus impairs or facilitates RTs when it is invalidly vs. validly presented alongside a trained or untrained (new) color, concluding that rewarded color captures attention when both stimuli have the same salience, and the cue is not predictive. In the same vein, Meyer et al. (2020) reproduced the results of Failing and Theeuwes (2014) but using a different training task, showing that the attentional capture effects of associating color to reward persist across different tasks.

The previous evidence could talk about the capacity of learned reward associations to produce effects on spatial cueing paradigms, although none of them brings light to the question of how reward interacts with exogenous and endogenous orienting. One study that tried to unmask that interaction is the one by Munneke et al. (2015) which used a spatial cueing paradigm with two different cues: a central predictive arrow (endogenous cue) and the change in the color of one of two placeholders with no predictive validity (exogenous cue). To manipulate reward, they associated three different colors to different levels of reward (high, low, and no reward) while participants performed the task (without a training phase). The main finding was the interaction between exogenous cueing and reward (larger cueing effect for highly rewarded color cues) independently of endogenous attention (i.e., central cue validity).

Considering the findings of Bucker and Theeuwes (2014) and Munneke et al. (2015) it is reasonable that different forms of reward delivery produce different effects in endogenous and exogenous orienting in a dissociated way. A recent study by Blini et al. (2018) has put that idea to test. In their study, Blini et al. (2018) used a setting like that of Bucker and Theeuwes (2014) and Munneke et al. (2015). They employ a shape discrimination task in a spatial cueing paradigm in which reward was provided on a trial-by-trial basis and was associated with the cue's color (high, low, and no reward). To prove the effects of color-reward associations on IOR, Blini et al. (2018) used an SOA of 700 ms between cue and target. Their results indicate that although a cue that predicted reward generally enhances performance, it did not modulate the IOR effect. From the perspective of Bucker and Theuuwes (2014), who considered IOR the result of endogenous orienting, this could imply that performance-based reward affects endogenous orienting while reward learning interacts with exogenous orienting.

However, the interpretation made by Bucker and Theeuwes (2014) of their results assumed that IOR is a consequence of, or related to, endogenous attention, although there is now ample evidence that IOR is not caused by the endogenous disengagement of attention from the previously attended location and the subsequent inhibition of the return of attention to it (Lupiáñez, 2010; Lupiáñez et al., 2013). Concretely, the assumption that IOR is attenuated by an enhancement of top-down control caused by reward-induced motivation to override IOR does not fit with evidence showing that IOR occurs even in situations in which attention is endogenously prompted to remain in the cued location? (Lupiáñez et al., 2004). Indeed, there is reasonable evidence to think that disengaging attention from the cued location is neither necessary nor sufficient to produce IOR (Chica et al., 2006; Chica & Lupiáñez, 2009).

In the same direction, the classic inhibition-of-return view would consider the processing of the cue, not the target, the critical aspect that is modulated by the motivation to affect the observed IOR effect, due to earlier or later disengagement of attention from the cue (Klein, 2000). However, other researchers rather consider IOR a consequence of the habituation of the orienting response to the target (Dukewich, 2009) or a reduced capacity of a cued target to capture attention, which leads to a detection cost due to cue-target integration processes (Lupiáñez et al., 2013). From both views, using a target with a strong capacity to capture attention could compensate for the habituation of the orienting response or reduce the detection cost, so that it would be possible to reduce or abolish IOR.

In line with this hypothesis, there is evidence showing that, compared with neutral targets, those with emotional valence could impair IOR. For example, Pérez-Dueñas et al. (2009) found that participants with high trait anxiety do not present IOR for negative words related to anxiety. Furthermore, Pérez-Dueñas et al. (2014) showed that IOR is abolished when faces with angry

emotional expressions are used as targets. In addition, Silvert and Funes (2016) discovered that faces with negative emotional expressions could impair IOR although only when facial processing is task-relevant (e.g., in emotional discrimination or gender discrimination tasks). However, it is not clear which factors of emotional faces or words beyond their high capacity to exogenously capture attention (Carretié, 2014) leads to the modulation of the IOR effect. By using essentially neutral stimuli, with no effective charge or intrinsic capacity to capture attention, and experimentally altering their capacity to exogenously capture attention (via reward learning; Failing & Theeuwes, 2014; Munneke et al., 2015) a much greater understanding of what factors may be involved in the expression of IOR would be gained. Nevertheless, to our knowledge, there is not a single study that has manipulated color-reward association over the target to prove if such manipulation could modulate IOR. Conducting such a study was the goal of the present research[1].

Therefore, the main objective of this preregistered study was to test whether and how color-reward association modulates the IOR effect in a spatial cueing paradigm with a letter discrimination task. Mainly, a procedure like that of (Pérez-Dueñas et al., 2014) was employed in which a discrimination task and two SOAs (100 ms and 1000 ms) are used. The main difference in design is that the task used involved discriminating between two letters (Z and M) that could be displayed in one of two colors (red and green), each of them associated (in a counterbalanced

[1] There was no study investigating this when we registered our first study (May 31st, 2021). However, after that we found out that two papers were published during the process of running our experiments, Li et al. (2021) and Wang et al. (2022). In the discussion, we will review these studies concerning the current study.

way) with one of two different levels of stimulus-reward contingency, giving rise to high-value and low-value target stimuli.

It is important to highlight different features of our design. We manipulated the color-reward association not in a pre-test phase, but "online" as in Munneke et al. (2015) and Blini et al. (2018). We used two levels of SOA (100 and 1000 ms) to respectively measure facilitation and IOR effects, and we wanted to test whether attentional bias to high-reward targets will persist even in the absence of reward, for which we added a final test without reward feedback. In addition, although we did not have strong predictions about the possible role of awareness over color-reward associations, we indirectly controlled any variability moderate by awareness. Concretely, participants were prompted to ignore color, but the reward structure was simple enough to realize which color was associated with each amount of reward (there were only two colors with two levels of reward, and the relation color-reward was completely deterministic). For that reason, we expected that most participants would be aware of the contingencies; nevertheless, we controlled this factor by using an awareness test at the end of the experiment.

Following our previous rationale, we predicted standard facilitation and IOR effects, except for the high reward stimulus, for which we expected a decrease and even complete abolition of IOR at the long SOA (like Pérez-Dueñas et al., 2014). In contrast, no differences in the facilitation effect as a function of reward association were expected at the short SOA, due to the high salience of the abrupt onset of cue and target and the fast integration processes occurring at the short SOA (Lupiáñez et al., 2013). Ultimately, we expected the above predictions to persist even in the absence of reward.

Experiment 1

Method

Transparency and Openness. The design, hypotheses and statistical analyses used in this study were pre-registered before data collection began (https://osf.io/fa6v9). All data, analysis code, and research materials are available at the following OSF repository.

Participants. The sample size calculation was carried out using the G*Power software Faul et al. (2007). However, there were no studies that had used a Posner-like task with manipulations regarding the color-reward association in the target. Some studies with a similar setting used here are those of Failing & Theeuwes (2014) and Munneke et al. (2015), in which they associated the color of a peripheral cue with different amounts of reward, the former finding that the effect size of the interaction between Reward and the use of an exogenous cue is η_P^2 = .301, and the latter finding an effect size of η_P^2 = .413 for the reward factor. On the other hand, the study by Pérez-Dueñas et al. (2014) shows that the target valence main factor effect size is η_P^2 = .779, and that of the interactive effects between cue and target valence for the long SOA (1000 ms) are of η_P^2 = .043, while Silvert & Funes (2016) find effects of η_P^2 = .22 (Experiment 3) and η_P^2 = .17 (Experiment 4) for the same interaction. Consequently, if we consider the above and add a recent meta-analysis on the ability of stimuli associated with rewards to capture attention when they are used as distractors (Rusz et al., 2020), given that it shows that the effect of this type of manipulation corresponds to a standardized mean change = .347, we decided to use an effect size of Cohen's *f* = .20. Thus, the a priori sample size was calculated for a 3 within-subject factor interaction (Reward x Validity x SOA), with 8 repeated measures (2 x 2 x 2), a value of α = .05, and using the default G*power option (correlation among repeated measures ρ = 0.5 and nonsphericity correction = 1) for the remaining values. With these settings, G*power yields that

between 24 (1 - β = 0.80) and 36 participants (1 - β = 0.95) are needed to have sufficient statistical power. Therefore, we decided to collect data from 40 participants.

The sample used consisted of undergraduate students at the University of Granada with normal or corrected vision and being naïve as to the purposes of the experiment. Participants were contacted through the distribution lists of the University of Granada, where those interested in participating in the experiment were asked to complete a brief survey in which they were asked to consent to the collection of personal data on general health variables and were administered the STAI-R test. Of those who passed the survey, 40 participants (M_{age} = 23.44, SD_{age} = 4.97; 30 women) were selected with scores between 15 and 34 on the STAI-R test (15th and 80th percentile according to Pacheco-Unguetti et al. (2010) in a sample of similar characteristics) and who did not show neurological or psychopathological conditions that could affect the purposes of the experiment (depression, ADHD or substance use; Anderson, 2020). Participants in the experiment could receive up to 10€, depending on their performance on the task (M = 9.77€, SD = .29€). Final data sample was collected in 2021.

Apparatus and Stimuli. The procedure and stimuli used were like those of Pérez-Dueñas et al. (2014), except that different targets were used. Specifically, the target consisted in one of two letters (Z or M; 60 pixels), which could appear in two different colors: red (RGB: 255, 0, 0) and green (RGB: 0, 255, 0). The targets appeared in one of two possible placeholders (128 x 128 pixels) to the left and to the right from a fixation cross (30 x 30 pixels), with 173 pixels in between. The fixation cross and the placeholder were painted in grey (RGB: 128, 128, 128), in a black (RGB: 255, 255, 255) background. Furthermore, for the cue, the placeholder changed briefly to a white (RGB: 0, 0, 0) and bolder (a 5-pixel increase in width) color. OpenSesame (Mathôt & Theeuwes, 2014) was used to program and control stimulus presentation and data

collection and the experiment was administered online, using the JATOS server (Lange et al., 2015) hosted by the Centre for Research, Mind, Brain, and Behavior (CIMCYC). Since other studies have already found reward effects online (Le Pelley et al., 2022; Watson et al., 2020), we do not expect that having conducted the study online was a limitation.

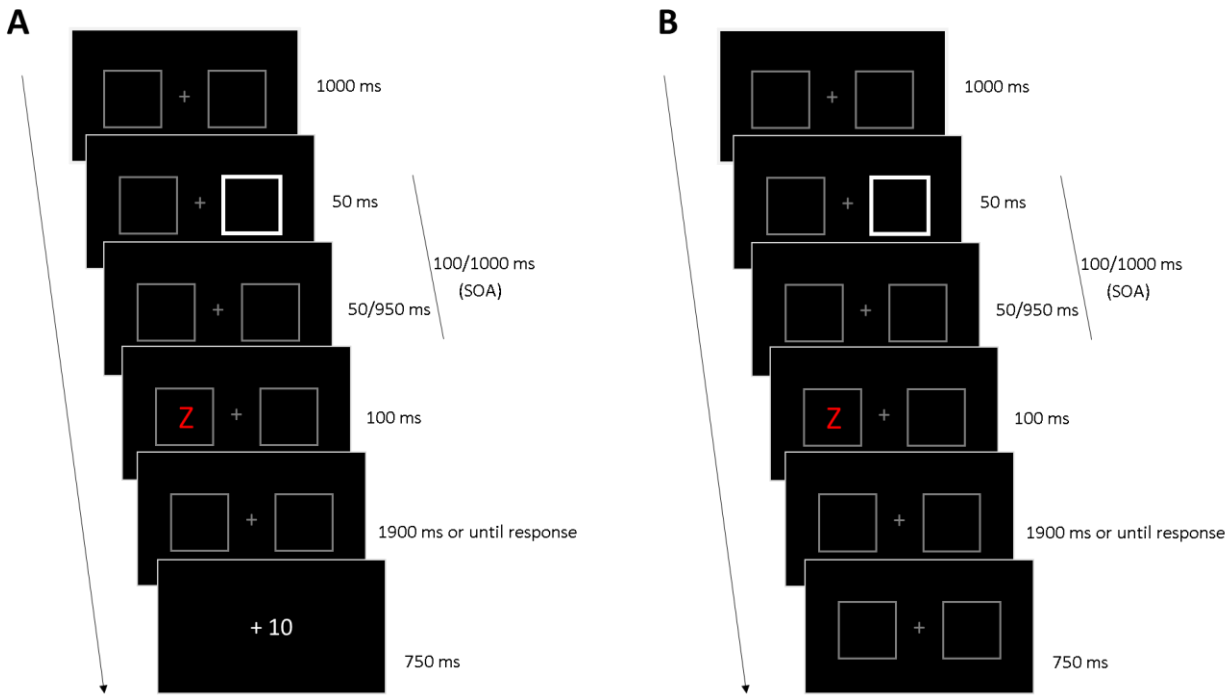
Procedure. Participants were asked to perform the task in a dimly lit room as quietly as possible to avoid distractions. At the beginning of the experiment, it was explained to the participants that their task was to discriminate two different letters (Z and M) using the letters "Z" and "M" on the keyboard while keeping their gaze on the fixation cross in the center of the screen. In addition, explicit instructions were given that throughout the experiment it was possible to earn points based on the stimulus displayed on the screen, which would be transformed into the accumulated real money they would receive at the end of the experiment. Participants were not warned of the specific color-reward association beyond what was presented in the feedback during the experimental phase.

As shown in figure 1, each trial started with a fixation cross in the center of the screen with two placeholders to the left and the right on a black background presented for 1000 ± 250 ms. Subsequently, one of the placeholders changed its color and width, acting as an attentional cue, which can occur with a 50% probability on each side of the screen during 50 ms. Afterwards, the change in illumination ceased, and the screen remained the same for 50 ms (short SOA) or 950 ms (long SOA). Immediately, one of the letters could appear on the screen for 100 ms, with equal probability on both sides of the screen, therefore 50% of the time it could appear at the place where the cue appeared (cued location trial), and 50% of the times at the opposite location (uncued location trial). After the appearance of the stimulus, it was possible to respond for up to a maximum of 1900 ms. Critically, the target could be displayed in two colors with

different reward levels: high-value (+10 points) and low-value (+1 point). The association of each color to high/low reward was counterbalanced across participants. Participants scored points for correct responses; however, incorrect or RT responses above 1200 ms resulted in the loss of the points earned (-10 points or -1 points). Finally, a counter would appear on the screen for 750 ms with the number of points gained or lost in that trial. The total amount of points and money gained was offered to the participant at the end of each block. The experimental sequence can be seen in Figure 1.

Figure 1

Experimental procedure



Note. The experimental task consisted in discriminating the shape of a letter that could appear on a left or right placeholder. Before the letter appears, one of the placeholders changes its luminance acting as an exogenous cue. In the reward phase (A), participants received a reward based on the color of the target. In the no-reward phase (B) the color-reward association was omitted, and participants were explicitly informed that there would be no more rewards.

The experiment started with a small practice block of 20 trials. In this block, participants received feedback on whether their answers were correct or incorrect and on their speed of response. Responses with RT below 200 ms received the message “TOO FAST”, while responses with RT above 1200 ms received the message “TOO SLOW”. If participants did not reach the criterion of 75% accuracy, they had to repeat the practice block. After the practice block, participants started the experimental block, in which they performed 8 blocks of 64 trials (512 trials) between which participants were allowed to see the total amounts of points and money earned and then they could rest. After finishing these 8 blocks, participants were given explicit instructions that there would be no more rewards from then on, and they had to perform 2 more blocks of 64 trials without financial rewards (128 trials) in which feedback was omitted and replaced with a 750 ms inter-trial screen with the fixation cross and the two placeholders.

Data analysis. Data were analyzed with R, version 4.1.2 (R Core Team, 2022), using the afex package (Singmann et al., 2022) for analysis of variance (ANOVAs), the emmeans package (Lenth, 2022) for contrasts and the BayesFactor package (Morey & Rouder, 2022) for Bayes Factor calculation. Following the analysis plan included in the pre-registration, the main DV used in this study was the mean RTs, but, in the case of detecting speed-accuracy trade-off, we intended to use instead the inverse efficiency score (IES; Inverse Efficiency score = mean RT / % hits), which allows us to account for variability in RTs as a function of participants’ accuracy (Bruyer & Brysbaert, 2011), having been successfully employed in other studies that have manipulated color-reward associations as a strategy to avoid the aforementioned speed-accuracy trade-off (Anderson & Halpern, 2017; Bucker & Theeuwes, 2017). For the RT analyses, incorrect responses (2.33 % in the reward phase and 3.59 % in the non-reward phase) and responses whose RT did not fall within the 200 - 1200 ms time window (0.43 % in the reward phase and 1.39 % in the non-reward phase) were discarded. In addition, as stated in the pre-registration, each of the

experimental phases (reward and non-reward) were analyzed separately, and we decided to calculate the bayes factor (BF_{01}) to check for evidence of absence in cases of evidence of absence, following the guidelines of Doorn et al. (2021).

Results

Reward Phase. To analyze the impact of reward during the first 8 blocks of trials (Figure 2A) we conducted a repeated measures ANOVA (2 x 2 x 2), with SOA (short and long), Value (high and low), and Cueing (cued and uncued). Our results shows a main effect of SOA, $F(1,39) = 31.39$, $p < .001$, $\hat{\eta}_p^2 = .446$, a main effect of Value, $F(1,39) = 6.51$, $p = .015$, $\hat{\eta}_p^2 = .143$, and a significative SOA x Cueing interaction, $F(1,39) = 46.62$, $p < .001$, $\hat{\eta}_p^2 = .544$. Such interaction reflect the above mentioned prediction of facilitation at the short SOA, $\Delta M = 12.96$, 95% CI $[8.52, \infty]$, $t(39) = 4.91$, $p < .001$, and IOR at the long SOA, $\Delta M = -18.04$, 95% CI $[-\infty, -12.12]$, $t(39) = -5.14$, $p < .001$. No more main effects or interactions were significant (*p*s > 0.24). The same analysis was repeated for Accuracy, showing no significant effects (*p*s > 0.34) indicating that the results were not influenced by speed-accuracy trade-off.

Given the possibility that the expression of the learned value effect may not have developed until later blocks (Failing & Theeuwes, 2014), we decided to test how the value effect varied across the 8 blocks of trials. To do so, we used a new repeated measures ANOVA with Value and Block as factors. Our results reveal significant effects of Value, $F(1,39) = 7.73$, $p = .008$, $\hat{\eta}_p^2 = .165$, Block, $F(3.47,135.52) = 3.78$, $p = .009$, $\hat{\eta}_p^2 = .088$, and a significant Value x Block interaction, $F(5.49,214.22) = 2.45$, $p = .030$, $\hat{\eta}_p^2 = .059$. Pairwise comparisons between high and low value targets shows that only in the last block were there significant differences,

$\Delta M = -15.96$, 95% CI $[-25.73, -6.19]$, $t(39) = -3.30$, $p = .002$. We repeated the previus analysis for the last block of trials and we find a main effect of SOA $F(1,39) = 28.51$, $p < .001$, $\hat{\eta}_p^2 = .422$, Value, $F(1,39) = 7.23$, $p = .010$, $\hat{\eta}_p^2 = .156$, a SOA x Cueing interaction, $F(1,39) = 6.10$, $p = .018$, $\hat{\eta}_p^2 = .135$, and a Value x Cueing interaction, $F(1,39) = 10.27$, $p = .003$, $\hat{\eta}_p^2 = .208$. Further analisys of the former interaction shows that while there was a significant value effect for uncued locations, $\Delta M = -23.32$, 95% CI $[-35.56, -11.08]$, $t(39) = -3.85$, $p < .001$, there was no value effect for cued locations $\Delta M = -5.40$, 95% CI $[-17.54, 6.74]$, $t(39) = -0.90$, $p = .374$. Besides, the three-way interaction remains not significant (p = .492). Importantly, there was no modulation of value over the IOR effect (i.e., Cueing x Value interaction) at the long SOA, neither overall, $F(1,39) = 0.00, p = .950$, BF_{excl} = 3.64, nor at the last block of trials, $F(1,39) = 0.48$, $p = .492$ BF_{excl} = 3.38.

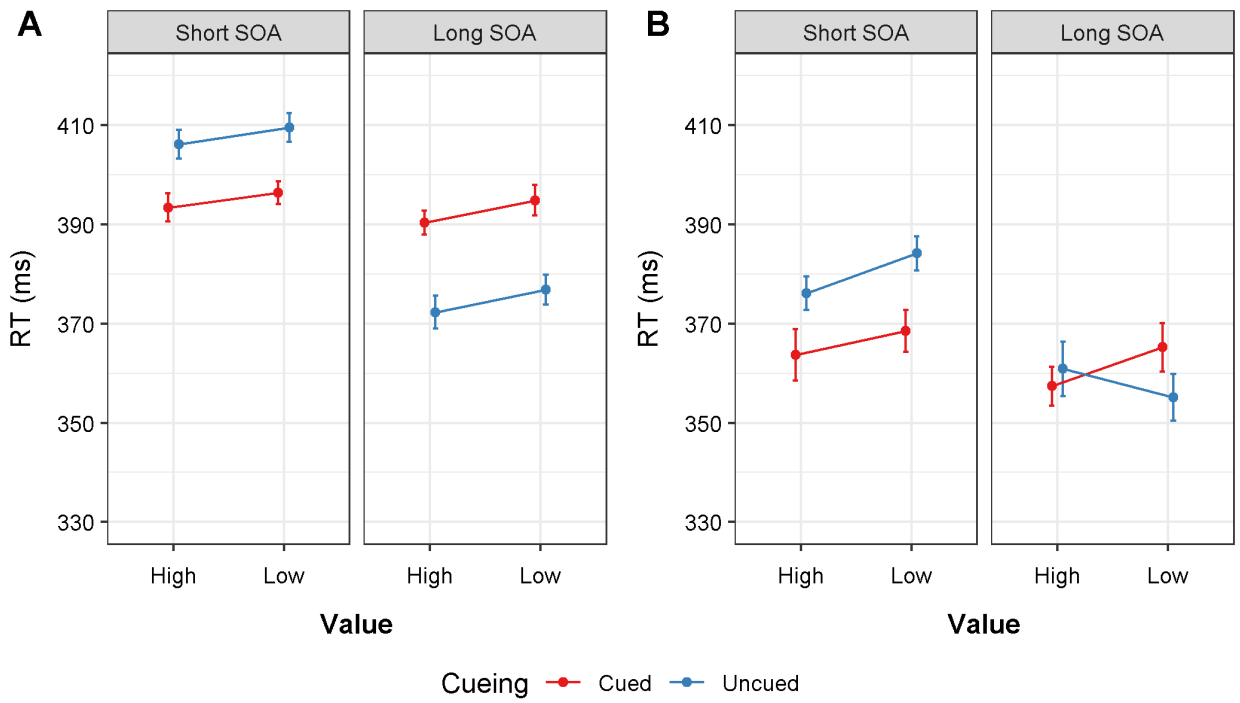
No-reward Phase. We conducted another SOA x Value x Cueing ANOVA for the two blocks of trials of the no-reward phase (Figure 2B). Such analysis reveals a main effect of SOA $F(1,39) = 9.03$, $p = .005$, $\hat{\eta}_p^2 = .188$, a marginal main effect of Value $F(1,39) = 4.03$, $p = .052$, $\hat{\eta}_p^2 = .094$, a significant SOA x Cueing interaction $F(1,39) = 9.13$, $p = .004$, $\hat{\eta}_p^2 = .190$, which shows facilitation for the short SOA, $\Delta M = 14.03$, 95% CI $[7.55, \infty]$, $t(39) = 3.65$, $p < .001$, but no IOR for the long SOA, $\Delta M = -3.31$, 95% CI $[-\infty, 5.13]$, $t(39) = -0.66$, $p = .256$, and a close to significance three-way interaction $F(1,39) = 3.31$, $p = .077$, $\hat{\eta}_p^2 = .078$, with no more significant interactions (ps > 0.13).

Although non-significant, given our a priori predictions and that there was no overall IOR effect, we decided to analyze this three-way interaction performing one ANOVA with Value and Cueing as factors for each level of SOA. Firstly, at the short SOA we found a marginally significant main effect of Value, $F(1,39) = 3.74$, $p = .060$, $\hat{\eta}_p^2 = .088$, and a significant main

effect of Cueing $F(1,39) = 13.31$, $p = .001$, $\hat{\eta}_p^2 = .254$, but no Value x Cueing interaction (p = .669). Furthermore, at the long SOA there was no main effect of Value or Cueing (*p*s > .512), but a significant Value x Cueing interaction $F(1,39) = 4.12$, $p = .049$, $\hat{\eta}_p^2 = .095$. This interaction arose from a significant IOR effect for the low-value target, $\Delta M = -10.12$, 95% CI $[-\infty, -0.64]$, $t(39) = -1.80$, $p = .040$, but no IOR effect for the high-value target, $\Delta M = 3.49$, 95% CI $[-\infty, 14.28]$, $t(39) = 0.54$, $p = .705$.

Figure 2

Plot for the results of Experiment 1



Note. A) Mean reaction times for each cueing (Cued and Uncued) condition, target type (High or Low value), and SOA level (Short or Long) in the reward phase. B) The same results, but in the no-reward phase. Error bars show Confidence Intervals using the Cousenau-Morey method (Morey, 2008).

As in the reward phase, we conducted another ANOVA with SOA, Value, and Cueing with the percentage of errors. The results show that none of the main or two-way effects were significant (*p*s > 0.16). However, the interaction between the three factors was significant $F(1,39) = 5.27$, $p = .027$, $\hat{\eta}_p^2 = .119$. As such results could imply that previous analyses may

be contaminated by speed-accuracy trade-off, according to the pre-registered analysis plan, we repeated the analyses employing the IES index as the DV.

Using the IES index instead, the results from the same ANOVA analysis shows a significant main effect for SOA,$F(1,39) = 12.06$, $p = .001$, $\hat{\eta}_p^2 = .236$, and Value $F(1,39) = 4.66$, $p = .037$, $\hat{\eta}_p^2 = .107$, a significant SOA x Cueing interaction, $F(1,39) = 7.38$, $p = .010$, $\hat{\eta}_p^2 = .159$, with facilitation at the short, $\Delta M = 15.00$, 95% CI $[6.78, \infty]$, $t(39) = 3.08$, $p = .002$, and no IOR at the long SOA, $\Delta M = -3.12$, 95% CI $[-\infty, 5.73]$, $t(39) = -0.59$, $p = .278$, and a significant three-way interaction, $F(1,39) = 7.45$, $p = .009$, $\hat{\eta}_p^2 = .160$.

To further explore this interaction, we run an ANOVA with Value and Cueing as factors for each level of SOA. Regarding the short SOA, we only found a main effect of Cueing, $F(1,39) = 9.46$, $p = .004$, $\hat{\eta}_p^2 = .195$, but no a main effect of Value nor a Value x Cueing interaction (*p*s > 0.12). In contrast, at the long SOA the expected Value x Cueing interaction was significant, $F(1,39) = 5.48$, $p = .024$, $\hat{\eta}_p^2 = .123$. As before, this interaction seems to be a consequence of the absence of IOR for the high-value target, $\Delta M = 6.77$, 95% CI $[\infty,\ 14.09]$, $t(39) = -0.92$, $p = .82$, and a significant IOR effect for the low-value target, $\Delta M = -13.01$, 95% CI $[6.9, \infty]$, $t(39) = 2.13$, $p = .02$.

Discussion

The main purpose of the study was to evaluate the potential ability of neutral stimuli contingently associated with economic rewards to modulate IOR. For this purpose, a study was carried out using a Posner-type task in which we varied the amount of economic reward that could be obtained depending on the color of the target. The results for the two phases indicate that the manipulations referring to value learning and the biphasic attentional orienting pattern as

a function of cueing and SOA have been successful, with facilitation effects for the high-value target, and by the presence of the interaction between SOA and cue type, with a facilitation effect at the short and IOR at the long SOA. Nevertheless, and contrary to our hypotheses, in the reward phase the target type did not modulate the interaction between SOA and Cueing, since the expected interaction between SOA, Value and Cueing was not significant. However, this interaction did prove significant in the no-reward phase, noting that while the target type did not influence the cueing effect at the short SOA, the predicted interaction was observed at the long SOA, with IOR being exclusively observed for the low-value stimulus. Thus, our predictions were fulfilled with the exception that we expected to observe such effects in both phases, whereas we only observed them when we omitted the possibility of obtaining rewards.

One possible explanation for why we did not observe any modulation of value over the orienting effects in the reward blocks may have to do with how the value manipulation itself was implemented in this experiment. Looking at other studies that have tried to observe the effects of value while rewarding performance, one thing they seem to share is that reward administration used to be within more restrictive time windows than those we employed (Munneke et al., 2015). On the other hand, the present pattern of results may derive from differences in the explicitness of the reward manipulation, with the color in the last two blocks being truly irrelevant, and this may have been the factor that allowed value learning to modulate the IOR effect. To test both possibilities, we decided to conduct a second experiment in which we manipulated the time window in which each participant could earn a reward.

Experiment 2

The main goal of this second pre-registered experiment was to explore why we did not find the modulation of IOR in the reward phase. The procedure was the same as experiment 1, except that we used the first two blocks of trials to compute the standard deviation in reaction time for each participant. In this way, we imposed an upper limit on the reaction times consisting of the mean of the reaction times plus one standard deviation. Furthermore, we increased the number of blocks of the no-reward phase to 4 instead of 2.

Method

Transparency and Openness. This second experiment was also pre-registered. The material related to the experiment can be obtained at the following link: https://osf.io/r2yf5. All data, analysis code, and research materials are available at the following OSF repository.

Participants. For this second experiment, we decided to increase the sample from 40 to 60 participants, to try to enhance statistical power. The participants, as in the previous experiment, are students from the University of Granada who were contacted through the mailing lists. Participants were selected based on the eligibility criteria described in the previous experiment. The final sample consisted of 60 participants *(M_{age}* = 22.22, *SD_{age}* = 3.35; 49 women). As in the previous experiment, participants could receive up to 10 €, depending on their performance on the task (M = 7.55€, SD = 1.16€), and the final sample was collected by 2021.

Apparatus and Stimuli. The experimental setup was the same as in experiment 1.

Procedure. The only two changes were that we used the first two blocks to calculate the standard deviation of each participant and put an upper limit on the response time to get a reward (+1*SD*), and we increased the number of blocks in the no-reward phase to four.

Data analysis. The data analysis will follow the same steps as in the previous experiment. In this case, for the RT analyses we excluded 7.66% of incorrect responses and 3.41% of responses that did not fall within the 200 - 1200 ms time window. The two blocks used to calculate the reward threshold were not included in the analysis. The only extra consideration is that since in the previous experiment the no-reward phase had only two blocks, in the case of not finding the effect, to replicate the previous experiment we decided to repeat the analyses for the first two blocks of trials.

Results

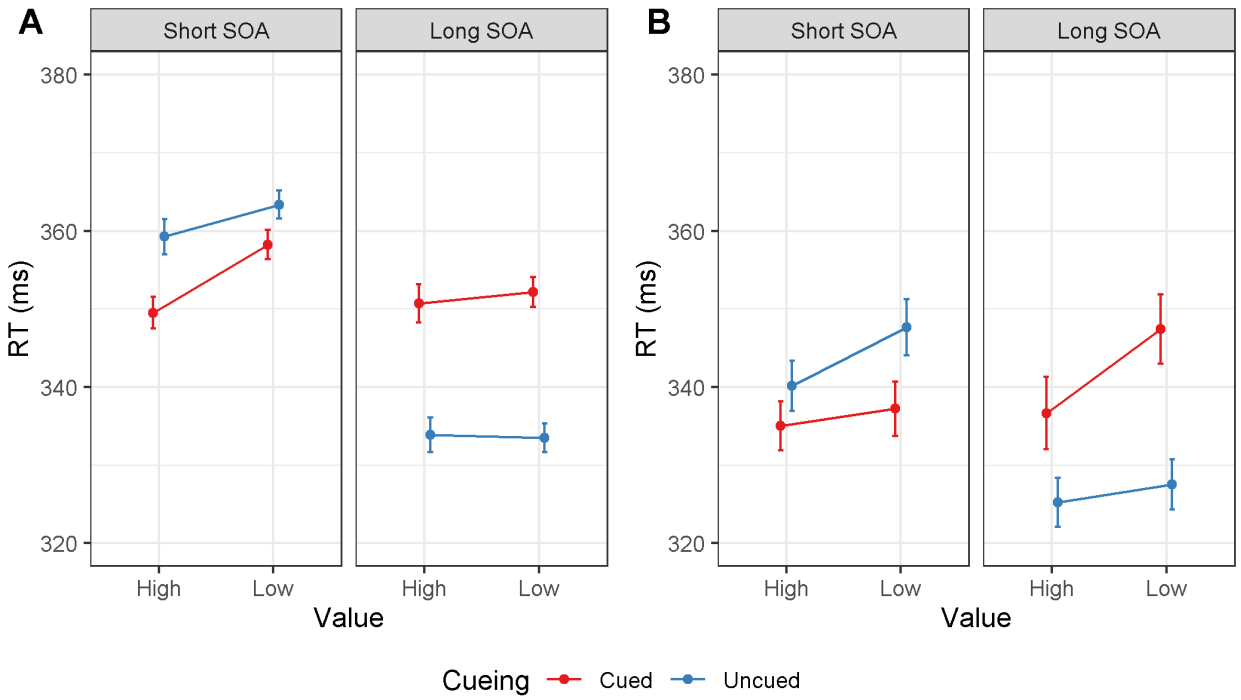
Reward Phase. As in the previous experiment, we conducted a repeated measures ANOVA (2 x 2 x 2), with SOA (short and long), Value (high and low), and Cueing (cued and uncued) for the reward phase (figure 3A). Our results shows a main effect of SOA, $F(1,59) = 47.73$, $p < .001$, $\hat{\eta}_p^2 = .447$, Value, $F(1,59) = 10.16$, $p = .002$, $\hat{\eta}_p^2 = .147$ and Cueing $F(1,59) = 8.86$, $p = .004$, $\hat{\eta}_p^2 = .131$. Furthermore, we found a significant SOA x Cueing interaction $F(1,59) = 55.03$, $p < .001$, $\hat{\eta}_p^2 = .483$, reflecting facilitation at the short ($\Delta M = 7.44$, 95% CI $[4.05, \infty]$, $t(59) = 3.67$, $p < .001$) and IOR at the long SOA ($\Delta M = -17.78$, 95% CI $[-\infty, -13.14]$, $t(59) = -6.40$, $p < .001$), but also a significant SOA x Value interaction $F(1,59) = 10.08$, $p = .002$, $\hat{\eta}_p^2 = .146$. This last interaction was due to an increased effect of value at the short, compared to the long SOA $\Delta M = 4.85$, 95% CI $[0.43,9.27]$, $t(59) = 2.20$, $p = .032$. There were no more significant main effects or interactions (ps > 0.15).

In the absence of a significant triple interaction, to test whether the effect has started to be expressed in later blocks, we performed an ANOVA with Block and Value as factors, finding main effects of block, $F(3.80,224.06) = 11.41$, $p < .001$, $\hat{\eta}_p^2 = .162$, and value, $F(1,59) =$

7.68, $p = .007$, $\hat{\eta}_p^2 = .115$, but no interaction between them (p = .105), suggesting that the manipulation has increased the effect of value throughout the experiment. Importantly, there was no modulation of value over the IOR effect at the long SOA, neither overall, $F(1, 59) = 0.54$, $p = .464$, BF_{excl} = 4.12, nor at the last block of trials, $F(1, 59) = 0.54$, $p = .891$, BF_{excl} = 6.53.

Figure 3

Plot for the results of experiment 2



Note. Plot for the results of experiment 2. A) Mean reaction times for each cueing (Cued and Uncued) condition, target type (High or Low value), and SOA level (Short or Long) in the reward phase. B) The same results, but in the two first blocks of the no-reward phase. Error bars show Confidence Intervals using the Cousenau-Morey method (Morey, 2008).

We repeated the analysis for the proportions of errors, finding a main effect of SOA, $F(1,59) = 9.35$, $p = .003$, $\hat{\eta}_p^2 = .137$, with a higher proportion of errors at the short ($M = 0.068$, 95% CI $[0.056, 0.080]$) than at the long SOA ($M = 0.060$, 95% CI $[0.047, 0.065]$). The main effect of Value was also significant, $F(1,59) = 6.97$, $p = .011$, $\hat{\eta}_p^2 = .106$, with more errors for high valued-targets ($M = .0654$, 95% CI $[.055, .076]$) than low valued ones ($M =$

.058, 95% CI [.049, .068]); and a close to significance effect of Cueing, $F(1,59) = 3.81$, $p = .056$, $\hat{\eta}_p^2 = .061$, Uncued targets showing more errors ($M = .065$, 95% CI [.054, .077]) than Cued targets ($M = .060$, 95% CI [.050, .068]). There were no more significant effects or interactions (*p*s > 0.32). As we only found main effects, we do not consider that interactions are affected by speed-accuracy trade off.

No-reward Phase. The same analyses were carried out in the no-reward phase (Figure 3B). We found a main effect of SOA $F(1,59) = 25.35$, $p < .001$, $\hat{\eta}_p^2 = .301$, an almost significant effect of Value $F(1,59) = 3.68$, $p = .060$, $\hat{\eta}_p^2 = .059$, and a SOA x Cueing interaction, $F(1,59) = 55.03$, $p < .001$, $\hat{\eta}_p^2 = .483$, showing facilitation at the short ($\Delta M = 7.44$, 95% CI $[4.05, \infty]$, $t(59) = 3.67$, $p < .001$) and IOR at the long SOA ($\Delta M = -10.95$, 95% CI $[-\infty, -6.12]$, $t(59) = -3.79$, $p < .001$). There were no more significant results (*ps* > 0.29). Regarding the proportions of errors, there were only a close to significant effect of SOA $F(1,59) = 25.05$, $p < .001$, $\hat{\eta}_p^2 = .298$. As it does not affect any interaction with value, we interpret that the value of the interaction of interest is not contaminated by speed-accuracy trade off.

Given that the previous experiment used only two blocks of trials in the No-Reward Phase, to see whether we replicated the previous findings, we repeated the above analysis for the first two block of the no-reward phase (figure 4). This analysis yield a main effect of SOA, $F(1,59) = 25.35$, $p < .001$, $\hat{\eta}_p^2 = .301$, and Value, $F(1,59) = 3.68$, $p = .060$, $\hat{\eta}_p^2 = .059$, and a SOA x Cueing interaction, $F(1,59) = 12.29$, $p = .001$, $\hat{\eta}_p^2 = .172$, with facilitation at the short ($\Delta M = 7.78$, 95% CI $[0.33, \infty]$, $t(59) = 1.75$, $p = .043$) and a significant IOR effect at the long SOA ($\Delta M = -15.66$, 95% CI $[-\infty, -9.58]$, $t(59) = -4.30$, $p < .001$). Nevertheless, the triple interaction remained not significant (p = .103). A specific ANOVA conducted on the data from

the long SOA shows only a significant effect for Cueing, $F(1,59) = 18.52$, $p < .001$, $\hat{\eta}_p^2 = .239$, but no effect of Value or a Value x Cueing interaction (*p*s > .173). Lastly, there were no significant main effects or interactions with the proportion of errors (*p*s > 0.23).

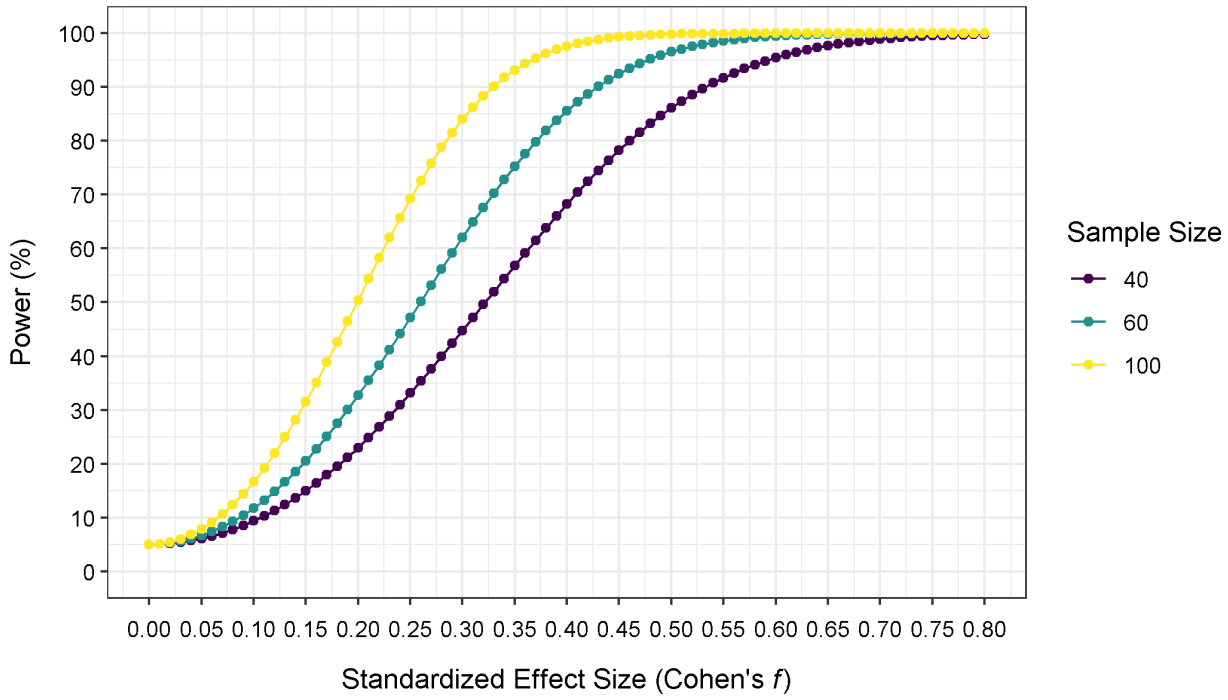
Discussion

In this experiment, we again showed that the modulation of the IOR effect does not occur in the reward phase, even after toughening the time window to gain the reward, as hypothesized. Moreover, we did not replicate the previous experiment, as expressed by the absence of a three-way interaction both for the entire non-reward phase and in the first two blocks (exactly as in experiment 1).

Although we can take these results as evidence for the inability of value-based attention to modulate the IOR effect, we must keep in mind that a null result does not mean that there is no effect (given experiment 1). Another plausible interpretation is that the effect we are looking for is smaller than we initially thought. It is widely known that the sample size needed to achieve good statistical power for interactions, especially over-additive interactions, is larger than for simple effects, and this sample size calculation also depends on the number of factors and the shape of the interaction (Lakens & Caldwell, 2021). In our sample size justification, we might have underestimated the sample size needed to achieve sufficient power. In fact, in this second experiment, there was a nominal reduction in the IOR effect for the high-value stimuli relative to the low-value stimuli ($\text{IOR}_{high} = -11.43$ and $\text{IOR}_{low} = -19.88$; figure 3B), although the Value x Cueing interaction was also non-significant for the long SOA.

Figure 4

Power–sensitivity plot



Note. Plots show power as a function of effect size for different sample sizes.

One way to examine the above line of reasoning is through a sensitivity power analysis (Lakens, 2022). Morey (2020) proposed a method for conducting a power analysis based on the relationship between effect size and power. Morey's method assesses the probability of a test statistic reaching the significance threshold as a function of the standardized effect size and the underlying distribution used to calculate the critical value of interest. This technique could be used to estimate the probability of observing a significant SOA x Value x Cueing interaction as a function of a range of effect sizes. This range of effect sizes should reliably represent the true effect size to be estimated. As we wanted to replicate the findings of the first experiment, we calculated a standardized effect size for the SOA x Value x Cueing interaction: Cohen's *f*. Cohen's *f* observed in experiment 1 was 0.29, 95% CI[. 0, .61], for RT and 0.44, 95% CI[. 1, .76], for IES. Figure 4 shows the Morey plot for the sample size for both experiments over the range of

possible effect sizes for the interaction. As can be seen, if we wanted to reach the standard power of 80%, for a sample size of 40, we would need a Cohen's f = .46, and for a sample size of 60, we would need at least a Cohen's f = .37.

Considering precision in our estimates and power, some authors have recommended that, when certain conditions are met, it helps to analyze different data sets to improve the estimation. In this regard, Curran and Hussong (2009) state that when several studies with original data are available, a pooled analysis of several data sets can help to improve power and assess replicability, while considering differences between studies, which seems to agree with other authors (Shrout and Rodgers, 2018). However, there are caveats to conducting a pooled analysis. For example, Vosgerau et al. (2019) argue that such a pooled analysis may increase power at the cost of increasing α. However, they agree that when all studies are included in the analysis, all experiments were pre-registered and therefore the analysis was planned, as in our case, it is feasible to perform this type of analysis.

Pooled analysis of Experiments 1 and 2

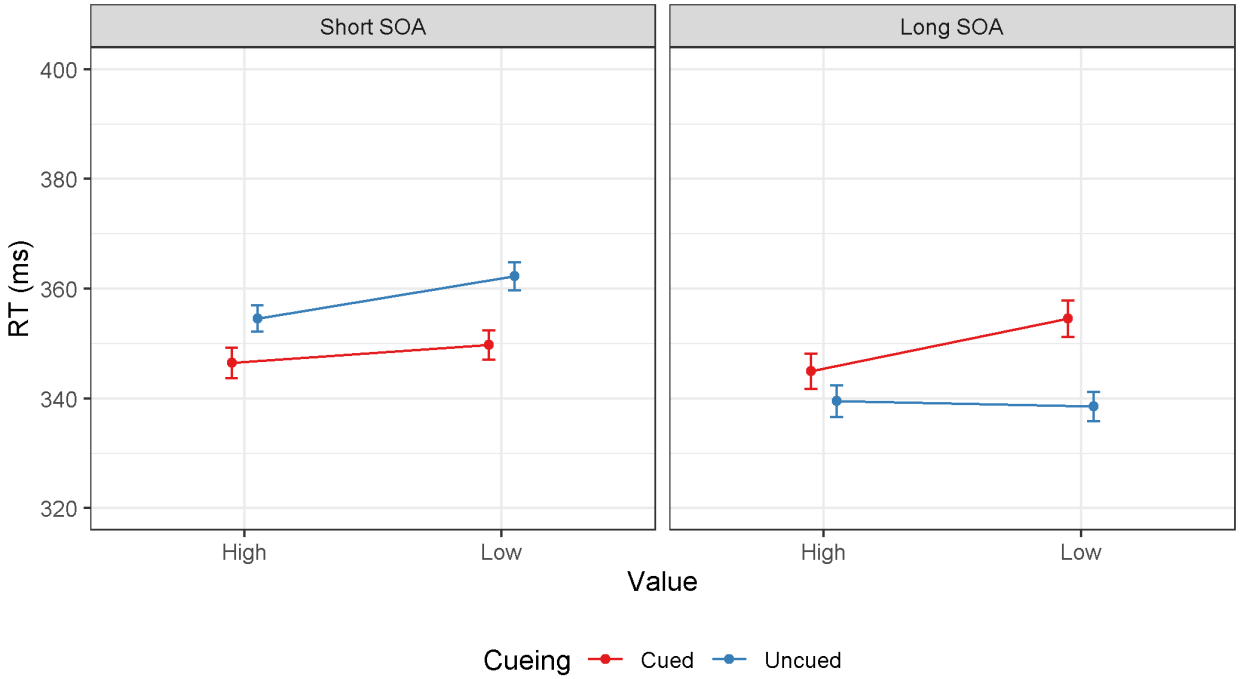
Considering the above, we decided to re-analyze the first two blocks of trials of the no-reward phase using the whole sample. Regarding previous analyses, we modified two things. The first is that accuracy analysis changed from an ANOVA over the proportion of error to a general linear mixed effect model (GLMM) over accuracy with a binomial distribution using the lme4 package (Bates, 2015). This change was driven after realizing that ANOVA over categorical data violates several ANOVA assumptions that could induce spurious effects (Jaeger, 2008). Furthermore, we included the Experiment as a between-participant factor in the ANOVA together with SOA, Value, and Cueing, as within-participant factors.

Regarding RT analysis, as shown in Figure 5, there was a main effect of experiment, $F(1,98) = 8.48$, $p = .004$, $\hat{\eta}_p^2 = .080$, due to RT being slower in experiment 1 ($M = 366.39$, 95% CI $[350.93, 381.85]$) than in experiment 2 ($M = 337.10$, 95% CI $[324.48, 349.73]$. A main effect of SOA, $F(1,98) = 15.26$, $p < .001$, $\hat{\eta}_p^2 = .135$ and a main effect of Value, $F(1,98) = 6.60$, $p = .012$, $\hat{\eta}_p^2 = .063$, were also observed. Furthermore, the Experiment x Cueing interaction was also significant, $F(1,98) = 5.10$, $p = .026$, $\hat{\eta}_p^2 = .050$, which seems to be related to an increased overall Cueing effect in experiment 2 ($\Delta M = 9.30$, 95% CI $[17.46,1.13]$, $t(98) = 2.26$, $p = .026$). Similarly, the SOA x Cueing interaction was also significant, $F(1,98) = 19.60$, $p < .001$, $\hat{\eta}_p^2 = .167$, with facilitation at the short ($\Delta M = 10.90$, 95% CI $[4.65, \infty]$, $t(98) = 3.46$, $p < .001$) and IOR at the long SOA ($\Delta M = -9.49$, 95% CI $[-\infty, -3.48]$, $t(98) = -3.14$, $p = .001$). Finally, and more importantly, the SOA x Value x Cueing interaction was significant $F(1,98) = 5.81$, $p = .018$, $\hat{\eta}_p^2 = .056$.

To disentangle this complex interaction, as above, we decomposed the interaction for each level of SOA. For the short SOA, there is a main effect of Value, $F(1,99) = 7.71$, $p = .007$, $\hat{\eta}_p^2 = .072$ and Cueing $F(1,99) = 11.09$, $p = .001$, $\hat{\eta}_p^2 = .101$, but no Value x Cueing interaction (p = .292). On the other hand, at the long SOA condition we find a main effect of Cueing $F(1,99) = 12.69$, $p = .001$, $\hat{\eta}_p^2 = .114$, and a Value x Cueing interaction, $F(1,99) = 4.84$, $p = .030$, $\hat{\eta}_p^2 = .047$. This Value x Cueing interaction seems to arise from the reduced and not significant IOR effect observed for high valued targets ($\Delta M = -5.46$, 95% CI $[-\infty, 1.08]$, $t(99) = -1.39$, $p = .084$) compared to higher and significant IOR effect observed for low valued targets ($\Delta M = -15.98$, 95% CI $[-\infty, -9.76]$, $t(99) = -4.27$, $p > .001$).

Figure 5

Experiments 1 and 2 combined analyses



Note. Mean reaction times for each cueing (Cued and Uncued) condition, target type (High or Low value), and SOA level (Short or Long) in the no-reward phase for both experiments 1 and 2. Error bars show Confidence Intervals using the Cousineau-Morey method (Morey, 2008).

For accuracy analysis, firstly, we fitted a logistic mixed model with the fixed effect of SOA, Value, Cueing, and Experiment. Additionally, we included Participants as a random effect, accounting for the variability in intercepts explained by Participants. The model shows a main effect of SOA ($\hat{\beta} = -0.19$, 95% CI $[-0.37, -0.01]$, $z = -2.06$, $p = .039$), with a lower proportion of errors in the long ($M = .033$, 95% CI $[.024, .042]$) than the short ($M = .037$, 95% CI $[.027, .047]$) SOA. Moreover, there is also a main effect of Experiment ($\hat{\beta} = 1.27$, 95% CI $[0.92, 1.62]$, $z = 7.06$, $p < .001$) showing a significant increasing in errors for Experiment 2 ($M = .037$, 95% CI $[.027, .047]$) compared to Experiment 1 ($M = .009$, 95% CI $[.004, .014]$). There were no more significant effects or interactions in the model, but the SOA x Value x

Cueing almost reached significance ($p = .074$). For that reason, we decided to re-analyze the data using the IES.

The analysis of the IES showed a significant effect of SOA $F(1,98) = 34.82$, $p < .001$, $\hat{\eta}_p^2 = .262$, a close to significant effect of Cueing, $F(1,98) = 3.61$, $p = .060$, $\hat{\eta}_p^2 = .036$ and a significant SOA x Cueing interaction $F(1,98) = 35.97$, $p < .001$, $\hat{\eta}_p^2 = .269$, with facilitation at the short ($\Delta M = 16.09$, 95% CI $[11.65, \infty]$, $t(98) = 6.03$, $p < .001$) and IOR at the long SOA ($\Delta M = -7.89$, 95% CI $[-\infty, -2.60]$, $t(98) = -2.48$, $p = .008$), but no SOA x Value x Cueing Interaction (p = .259) and no more significant effects (ps > 0.15).

Given that our hypotheses concerned the IOR effect at the long SOA, we decided to explore this non-significant interaction for each level of SOA. At the short SOA there was only a main effect of Cueing $F(1,99) = 39.21$, $p < .001$, $\hat{\eta}_p^2 = .284$. At the long SOA, however, there was a main effect of Cueing, $F(1,99) = 7.92$, $p = .006$, $\hat{\eta}_p^2 = .074$, and a Value x Cueing interaction $F(1,99) = 5.65$, $p = .019$, $\hat{\eta}_p^2 = .054$, reflecting no IOR for high-valued targets ($\Delta M = -2.77$, 95% CI $[-\infty, 4.15]$, $t(99) = -0.66$, $p = .254$), a larger and significant IOR for low-valued targets ($\Delta M = -14.92$, 95% CI $[-\infty, -8.39]$, $t(99) = -3.79$, $p < .001$).

General Discussion

The main purpose of these two pre-registered studies was to test whether IOR can be affected by the mere association of task-irrelevant dimensions in the target with economic rewards. To do so, we manipulated the amount of reward associated with the color in which the target could appear in a Posner-like task. In our first experiment, we found that the reward associated with the target color modulated the IOR effect, but only when feedback was omitted. In the second experiment, we failed to replicate this finding statistically, although we found the

pattern nominally. After a sensitivity analysis of the effect size, we decided to perform a pool analysis on the two experiments to increase the power in the estimation and found that the effect was maintained when we pooled the whole sample (N = 100).

If we try to explain why learned value can modulate the IOR, it is relevant to consider other studies in which it has been possible to modulate the IOR by manipulating the saliency of the target. As mentioned in the introduction, when emotional stimuli are used as targets, it appears that it is possible to modulate IOR when emotional processing is task-relevant, and this modulation of IOR has been attributed to increased attentional capture by the target. Similarly, learned value (or selection history) effects have typically been explained by increased attentional capture produced by stimuli associated with other appetitive stimuli. Moreover, this finding is even more revealing if we consider that Blini et al. (2018) did not observe any effect of the learned value on IOR when the manipulation is applied to the color of the cue. Therefore, it seems that regarding the influence of the learned value on IOR, what matters is the saliency of the target, not the cue. This fact contradicts the classical view of IOR, where the interpretation of the effect is produced by a bias against returning attention to a previously attended location (Klein, 2000) and supports theories that explain IOR as a decrease in the capture produced by a previously cued location (detection cost; Lupiáñez et al., 2013) or as an increase in habituation in the orienting response (Dukewich, 2009).

As we have shown, in all experiments there was a robust absence of IOR modulation due to target value in the reward phase. Although unexpected, this result can be explained for several reasons. First, if we look at the selection history literature, there are two main paradigms employed to produce this effect: Anderson's paradigm (Anderson, 2011), which uses two phases (learning and test), and Le Pelley's paradigm (Le Pelley, 2015), which uses only one. Both are

based on the additional singleton paradigm (Theeuwes, 1992), in which participants must attend to a stimulus that is relevantly unique in one feature (usually shape) and ignore any uniqueness of a different feature (usually color). In Anderson's paradigm participants are trained with a task before the test phase, where participants are presented with multiple colors, but only two of them can act as targets, one associated with a high reward, and the other with a low reward. Subsequently, colors that have previously served as targets are used as distractors in the same task.

On the other hand, in Le Pelley's paradigm, there is no prior training phase, and participants receive conditional feedback on the color of the distractor. Therefore, the effect emerges through an increasing exposure to color–reward contingencies. However, attending to the singleton distractor is counterproductive, as the reward is determined by performance. In this sense, in both paradigms color is irrelevant, as there is no real instrumental connection between color and reward when value modulates capture. In the reward phase of the present study there was an instrumental relationship between color and reward, so participants may have formed an attentional set for color, which may have resulted in color capturing attention endogenously. In contrast, in the no-reward phase, it is possible that color became a completely irrelevant feature, which may have led to bottom-up modulation of salience, a type of capture that is known to be amplified by learned value (Anderson et al., 2011b; Hinault et al., 2019).

Another possibility is that the observed results are not due to the manipulation per se, but to a practice-induced decrease in the IOR effect. In our view, this explanation seems unlikely. Specifically, according to Lupiáñez et al. (2001), increased practice with this spatial cueing paradigm is usually accompanied by a decrease in both IOR and facilitation effects. Although this decrease is more pronounced in discrimination tasks, they showed that it is largely attenuated

when the SOA is manipulated within blocks, as in the present study. On the other hand, if the results are driven by practice, the reduction in IOR should affect both the high and low-value targets and not a specific reduction in IOR for the high-value target.

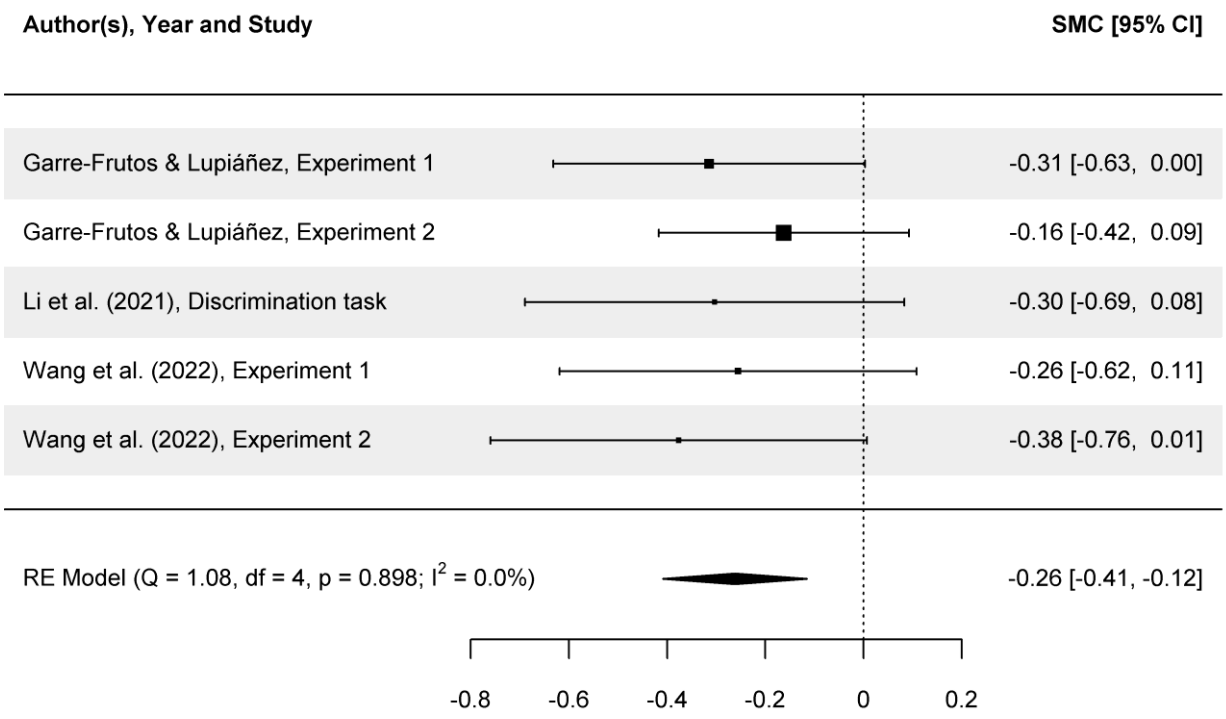
This study is the first to provide empirical evidence that it is possible to abolish the IOR effect by manipulating the value associated with an irrelevant dimension of the target, although it was not the first to attempt to show evidence of this possibility. After pre-registering the studies and in the process of carrying out the experiments and analyzing the data, two papers have been published that have attempted to study the same manipulation. First, Li et al. (2021) conducted an experiment in which they also used a Posner-like task to test the effects of learned value on IOR, with a single SOA (800 ~ 1000 ms) and two different tasks: a localization task and a color discrimination task. In both tasks, targets could be presented in two possible colors counterbalanced at two reward levels (high and low). However, unlike in our study, the associations were created in a pre-training task. Their results show that, at the behavioral level, both high-value and low-value targets show IOR. However, at the level of evoked potentials, they found that IOR markers were exclusively observed for the low-value target, not the high-value one. In fact, although in general both stimuli show IOR, nominally there was a reduction of the IOR effect in the discrimination task ($\text{IOR}_{high} = -7$ ms and $\text{IOR}_{low} = -16$ ms), which did not occur in the localization task ($\text{IOR}_{high} = -13$ ms and $\text{IOR}_{low} = -14$ ms). The former result converges with ours in a similar task. The fact that they did not find significant results on the modulation of IOR by reward may be due to statistical power issues ($N = 24$).

Second, Wang et al. (2022) conducted an experiment in which they found that targets associated with high reward reduces IOR in 3D space when the targets are presented in the near-depth plane. Their results are interesting, but the inference that could be drawn differs from that

of the present study because of how they manipulated the target value. According to Anderson (2021) one of the essential characteristics for us to consider an effect part of the selection story is that the effect is due to an empirically generated association to an arbitrary stimulus, but Wang et al. (2022) used real money as targets. Therefore, although the study by Wang et al. (2022) is informative, to the best of our knowledge, we believe our study is the first one to empirically demonstrate that it is possible to affect IOR through the association between an arbitrary stimulus feature and reward.

Figure 6

Meta-analysis on the size of the effect across different studies



Note. Forest plot showing the estimated standardized mean change of IOR between high and low/reward conditions. The left column shows the authors, year, and experiment numbers. The middle column shows the individual study effect sizes (black squares), with estimated 95% confidence intervals (error bars). The size of the square reflects the sample size of the study. The right column shows these effect sizes and 95% CIs. RE model random-effects model.

One of the major issues of this study is that, although we have gathered evidence to claim that reward-predicting stimuli that capture attention can attenuate IOR in the present work, the actual size of the effect may be smaller than initially thought. To estimate the potential effect size of the modulation of IOR by learned value, we conducted a meta-analysis of our two experiments conducted in the present study and the work of Li et al. (2021) and Wang et al. (2022), which are conceptually similar. We calculated the standardized mean change (*SMD*) of IOR in high and low-value conditions for long SOA. Figure 6 depicts the results of this analysis[2]. As can be seen, the size of the effect is significant across studies (*SMC* = -.262, 95% CI [-.408, -.116]) showing that IOR tends to be lower for high-reward predicting stimulus, and the test for heterogeneity shows no significant between-study variation ($Q(4) = 1.0771$, $p = .898$, $I^2 = 0.00\%$). We did not find evidence for publication bias (funnel asymmetry: $z = -0.8751$, $p = .381$; rank correlation test: $\tau = -.40$, $p = .483$), and even if assuming publication bias, the effect still holds significant (*SMC* = -.218, 95% CI [-.344, -.092). Based on this meta-analysis, to have at least 80% power to detect this effect size with a classic paired-*t* test, a sample size of at least 117 participants would be necessary. These results along with our previous sensitivity analysis further suggest that although there seems to exist an overall effect, each study seems to be severely underpowered to show it.

Giving the above, it is advisable to be cautious with the inferences drawn from the present study and all those published to date, as the actual effect size may well be rather small and short-

[2] For Experiments 1 and 2 of the present work, we employed the observed correlation between IOR for high and low-value targets (.38 and .09, respectively) to compute the standard errors of the SMD. As we cannot know the observed correlation of the other studies included in the meta-analysis, given that measures of differences as IOR usually suffer from measurement error (Draheim et al., 2019), we decide to assume a very low correlation based on Experiment 2 low correlation ($r = .10$), that may have been conservative regarding the real correlation in the studies by Li et al. (2021) and Wang et al. (2022).

lived, as we have noted in our meta-analysis. In psychology, there is currently a debate about the importance of small effect sizes. Advocating small effects, Götz et al. (2022) claim that most psychological phenomena are constructed based on multiple factors, each having a small influence. Yet researchers, for various reasons, are biased to focus on the search for large effect sizes and may ignore along the way effects that, although small, may well be determinant in describing the phenomena of interest. Against this line of reasoning, Primbs et al. (2022) point out that the above stance may encourage other researchers to judge any effect size as "indispensable" when one should also be critical given the shortcomings that Psychology harbors in terms of the measurement of its object of study. Regardless of the position, there is ample evidence that the association of an arbitrary feature of a stimulus with reward affects its ability to capture attention (Anderson, 2016; Failing & Theuwees, 2018; Anderson, 2021). Nevertheless, this influence may be more relevant in visual search paradigms where learned value is supposed to affect attentional allocation exogenously (Rusz et al., 2020). Therefore, we recommend that future researchers who intend to explore the relationship between the learned value of a target and IOR design their studies considering that a large sample size may be necessary to detect significant modulations of IOR by learned value.

In summary, this study aimed to investigate the effect of learned value on IOR in a Posner-like task. Results showed that the reward associated with the target color had a modulatory effect on IOR, but only when feedback was omitted. The results suggest that the influence of learned value on IOR is dependent on target salience and support theories explaining IOR as a decrease in the strength with which a previously cued target captures attention or an increase in the habituation of the orienting response. The study is the first to provide empirical evidence for the influence of learned value on IOR, although further research is needed to assess the robustness of the phenomenon.

Declaration of interests

The authors report there are no competing interests to declare.

Author contributions

Author roles were classified using the Contributor Role Taxonomy (CRediT; https://credit.niso.org/) as follows: *Francisco Garre-Frutos*: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Visualization, and Writing - original draft; *Juan Lupiáñez*: Conceptualization, Funding acquisition, Supervision, and Writing - review & editing.

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